

Lesson plan

Name of the Programme:	B.Tech (Hons.)	
Title of the Course:	Engineering Explorations	
Course code	CS1841	
Semester:	I/II	
Names of the Course Instructor(s)	Dr. Ramesh S	
Course credit	No. of hours per week (2+0+2)	Total no. of Teaching hours
03	04 Hours	60 Hours

Sessio n	Module No.	Topic	Pedagogy /activities	Pre-reading/referen ce
1	1	Introduction to Engineering Visualization	Lecture, Discussion on importance	Textbook 1: Ch.1
2		BIS Conventions of Engineering Drawing	Lecture, Examples, Chalk Talk	Textbook 1: Ch.2
3		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
4		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
5		Freehand Sketching	Sketching Activity	Textbook 1: Ch.3
6		Scales (Only Introduction)	Lecture, Demonstration	Textbook 1: Ch.4
7		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
8		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
9		Intro to CAD Software, Coordinate Systems	Software Demo, Hands-on Practice	Reference Book 1: Ch.1-2

10		Orthographic Projections: Points (1st & 3rd quadrant)	Whiteboard & CAD software practice	Textbook 1: Ch.6
11		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
12		Orthographic Projections of Solids	Lab/ Hand-on	Textbook 1: Ch.1
13		Orthographic Projections: Lines (1st quadrant)	Projection Exercises	Textbook 1: Ch.6
14		Orthographic Projections of Planes	Activity using Change of Position Method	Textbook 1: Ch.7
15	2	Isometric Projection of Solids	Lab/ Hand-on	Textbook 1: Ch.1
16		Isometric Projection of Solids	Lab/ Hand-on	Textbook 1: Ch.1
17		Projections of Solids: Prisms	CAD Drawing & Practice	Textbook 1: Ch.8
18		Projections of Solids: Pyramids	CAD Drawing & Practice	Textbook 1: Ch.8
19		Isometric Projection of Solids	Lab/ Hand-on	Textbook 3: Ch. 4
20		Isometric Projection of Solids	Lab/ Hand-on	Textbook 3: Ch. 4
21		Projection of Solids: Cube and Composite Solids	Software Practice	Textbook 1: Ch.8
22	3	Isometric Scale and Projection	Isometric Drawing Exercise	Textbook 1: Ch.9
23		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
24		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
25		Isometric Projection of Solids: Cube, Prisms	Isometric Sketching and CAD	Textbook 1: Ch.9
26		Isometric Projection: Cylinder, Cone, Sphere	Hands-on Drawing	Textbook 1: Ch.9

27		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
28		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
29		Isometric Projection of Solid Combinations	CAD Practice	Textbook 1: Ch.9
30		2D Drawing Project Planning	Team Activity, Brainstorming	Instructor Notes
31		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
32		Application of 2D drawing to Interdisciplinary Application	Lab/ Hand-on	Textbook 1: Ch.8
33		2D Drawing Project: Execution	Lab Work	Instructor Notes
34	4	Intro to Additive Manufacturing	Lecture, Video Demo	Textbook 2: Ch.1
35		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1
36		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1
37		Steps & Benefits of AM	Class Discussion, Mindmap	Textbook 2: Ch.2
38		AM Techniques and Materials	Poster Presentation	Textbook 2: Ch.3
39		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1
40		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1

41		3D Printing Workflow, Post-processing	Lecture + Case Study	Textbook 2: Ch.4
42		AM Applications + Quality Checking	Presentation and Discussion	Textbook 2: Ch.5
43		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1
44		Learn to create 3D models using solid-edge software and prepare them for printing using	Lab/ Hand-on	Textbook 2: Ch.1
45		Project: 3D Shape Reverse Engineering	CAD Lab Session	Reference Book 2
46	5	CAED for AM: Sketching & Modeling	Hands-on CAD, Demonstration	Reference Book 1: Ch.4
47		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	Textbook 2: Ch.1
48		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	Textbook 2: Ch.1
49		CAD Environment and File Formats	Group Discussion, File Exploration	Reference Book 1: Ch.5
50		Geometry Export/Import, Slicing	Live Demo in CAD Software	Reference Book 1: Ch.6
51		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	
52		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	
53		Layer Path, Part Orientation	Software Simulation	Reference Book 1: Ch.7

54		Project: 3D Printing Setup	3D Printer Hands-on	Reference Book 2
55		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	Textbook 2: Ch.1
56		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	Textbook 2: Ch.1
57		Project: Printing & Post-processing	Lab Activity	Reference Book 2
58		Final Presentation of 3D Projects	Group Presentation	Instructor Notes
59		Create the Engineering model by using reverse engineering Applications and printing the	Lab/ Hand-on	Textbook 2: Ch.1
60		Create the Engineering model by using reverse engineering Applications and printing the	Lab/Hands-on	Textbook 2: Ch.1

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Hand Sketch and Computer Sketch	20 marks out of 70
CIE-2 - Hand Sketch	25 marks out of 70
CIE-3 - Sketch book and Printouts submission - 3D printing Project Submission	15 marks out of 70 10 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Practical

Weightage of exam	30%

Course Outcomes- Program Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	Apply BIS conventions, freehand sketching, orthographic projections, and computer-aided drafting concepts to visualize and represent engineering components in 2D and 3D.	2	2	2	2	2	2	2	1	1	1	-	2
CO2	Construct orthographic projections of points, lines, planes, and solids using first-angle projection principles, demonstrating a clear understanding of spatial visualization	3	2	2	3	3	-	-	-	1	1	-	2
CO3	Evaluate isometric projections of various solids and their combinations using isometric scale and projection techniques to represent real-world 3D objects.	3	3	2	3	2	2	2	1	1	1	-	2
CO4	Demonstrate the use of CAED tools in additive manufacturing processes by creating, modeling, and 3D printing simple geometric shapes using proper slicing, orientation, and post-processing methods.	3	3	3	3	3	2	2	-	1	1	-	2

Rubrics for evaluation of CIE (Separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Manual sketch (10 Marks)	Marks will be distributed equally based on the Number of Positions.
	Computer Sketch (10 Marks)	Marks will be distributed equally based on the Number of Positions.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Only Manual Skecth	A test will be conducted for 25 marks, and the question paper pattern and Scheme of Evaluation will be prepared by a team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab record+ Viva = 10 marks 3D Printing Project Submission =15 marks (

Signature of the Course Faculty